Atmospheric entry

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

SUCCESS CRITERIA - I can...

1. Compare kinetic energy with a stated heating requirement.
2. Explain the role of surface area relative to mass.
3. Name missing factors in a survival prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ablation” and “thermal energy” to explain your model or result. Sentence starters are optional.

ablation: loss of surface material during heating or erosion

thermal energy: internal energy associated with microscopic motion/interactions

impact: a collision or strike between objects

01 Warm-up: bring back an earlier idea

A system receives 50 J heat and does 30 J work on its surroundings.
Find ΔU with $\Delta U = Q - W_{by}$.

Word help: surroundings = everything outside the chosen system

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

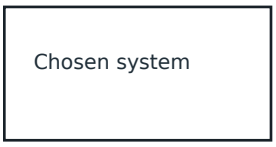
Reference: $K = \frac{1}{2}mv^2$
$Q_{absorbed}$ = fraction absorbed × energy available.
Entry also involves ablation and fragmentation. An energy comparison alone cannot predict survival.

Worked reasoning: A fictional entering object loses 1000 J of mechanical energy; a model assigns 700 J to heated air and 300 J to the object. Those shares are supplied assumptions, not universal atmospheric-entry fractions.

Your turn: Draw the energy budget and identify one measurement needed to test the shares.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Compare a small and a large sphere of the same material. Which has more surface area per unit mass? What does that suggest about surface interactions per kilogram?

Sentence starter: Both … . They differ in … because … .



A B

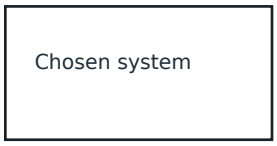
Marzano check - circle: 1 2 3

04 Apply the physics

An atmospheric-entry simulation assigns a model object 100 J of kinetic energy. A specified heating change requires 20 J.
Find the ratio, then draw two possible energy partitions. Does enough total energy prove that all of it heats the object?
Explain.

Sentence starter: My claim is … . The diagram/data show …, so … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

Name at least three additional quantities needed before claiming a particular rock completely ablates or reaches the ground.

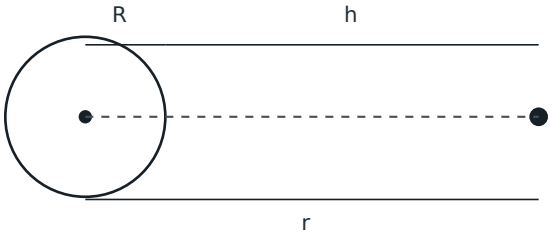
Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simplified model requires 24 J to escape, but an object has only 20 J and no later energy input. Does the model predict escape?
Explain.
Label the distances or energy levels used in your reasoning.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3

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Sentence starters are optional. You may plan in any language.

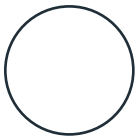
07 Exit check

Two equal-density spheres have radii r and 2 r. Supplied fact: area grows 4 times and mass 8 times.
Compare area per mass.
Label energy entering, leaving or stored on the system model.

Sentence starter: Both … . They differ in … because … .



Radius r



Radius 2r

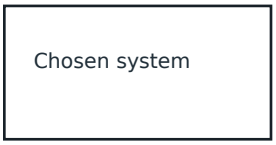
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Compare two objects with the same 1000 J energy loss but different assumed fractions heating the object: 20% and 60%.
Find both heating amounts and explain why survival is still uncertain 0.
Label energy entering, leaving or stored on the system model.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3